Housing First or Treatment First?

The model, its parameters, its calibration, and what the referees still want

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Abstract

A complete statement of the rebuilt model: environment, preferences, budget, every transition, the household problem, and the equilibrium. Section 3 lists *every* parameter with its symbol, its role, and either its external source or the moment that pins it. Sections 4–7 cover the two-stage calibration, the untargeted moments used for validation, and the mechanism decomposition that replaces the static appendix. Section 8 lists, referee by referee, every comment that is still outstanding.

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1 Overview

The model is a life-cycle economy in which households choose housing, drug use, treatment, hours and saving under mortality, employment, mental-health and addiction risk, and in which a housing voucher programme is run under one of two rules. The question is whether the voucher should be conditioned on abstinence.

Relative to the submitted paper the model has been rebuilt rather than patched. Homeownership, the housing market, tenure, mortgages and negative assets are gone; the drug price is now an equilibrium object; treatment exists; mental illness exists; preferences no longer vary with addiction status; labour supply is continuous; and every parameter is attached either to a citation or to a moment. Table 8 in Section 8 maps each change to the comments it answers.

Two conventions used throughout. Time is annual. All monetary quantities are in units of average model income, so "0.25" means a quarter of average income.

2 The model

2.1 Demographics and timing

Households enter at model age $j = 1$ (real age 26) and live at most to $j = J = 60$ (real age 85). They retire at $j_R = 41$ (real age 66). Survival from j to $j + 1$ is $\phi_j(h', d, m)$, defined in (13) below: it depends on age, on the housing chosen this period, on substance use disorder and on mental illness. A household that dies leaves its assets behind; the estate accrues to the government (Section 2.14).

Within a period the household observes its state, chooses hours, housing, drug use, treatment and saving, and consumes. Every shock — productivity, employment, addiction, mental health, voucher receipt and survival — is drawn *after* all of that and determines the state the household wakes up in at $j+1$. Nothing realised this period changes anything the household did this period, which is what makes those variables state variables rather than choices, and it is why the transition kernels in (11)–(14) condition on the CHOSEN h' and v: the choice comes first, the draw second.

2.2 The state

Two features are deliberate.

Housing entered with is not a state. Survival, employment transitions and the recovery hazard all condition on the *chosen* housing h', and renters face no adjustment cost, so the unit a household arrives in is payoff-irrelevant. Carrying it would replicate the whole problem N_h times; removing it leaves every aggregate unchanged and cuts the solve by a factor of 6.6. It would have to be reinstated if moving costs were added.

Three employment states, not four. UI is drawn on *entry* into unemployment with probability π_{UI} and lasts one year. A first-year non-recipient and a household unemployed for two or more years have identical income (zero) and identical continuation values, so those two states merge *exactly*. This answers the AE's A2: UI expiration matters, and it costs one state rather than two.

Table 1: The individual state: 12,960 points per age, 777,600 in total

	Symbol	Points	
Age	j	60	real age 26–85, retirement at 66
Financial assets	$a \geq 0$	60	cubic spacing, dense near zero; no borrowing
Labour productivity	y	9	Rouwenhorst discretisation of an AR(1) in logs
Employment	e	3	employed; unemployed *with* UI; unemployed without UI
Substance use disorder	d	2	$d = 0$ clean, $d = 1$ SUD
Serious mental illness	m	2	$m = 0$ absent, $m = 1$ present
Housing voucher	η	2	$\eta = 0$ not held, $\eta = 1$ held

2.3 Choices

Each period a household chooses

$$\big(n,\ h',\ v,\ \tau,\ a'\big) \in [0,1] \times \{h_1, \ldots, h_4\} \times \mathbb{R}_+ \times \{0,1\} \times \mathcal{A},$$

hours n, housing h' (where h_1 denotes homelessness and $h_2 < h_3 < h_4$ are rental sizes), drug consumption v, treatment take-up τ (available only if $d = 1$), and next-period assets a' on the asset grid.

2.4 Preferences

Period utility is

$$u = \frac{\big[c^{\alpha}\,(h')^{\beta}\,(v + b_{dmh'})^{\gamma}\big]^{1-\sigma}}{1-\sigma} \;+\; \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} \;+\; \aleph \;-\; \tau\,\bar{\tau}^{u} \;-\; \kappa_x\big(H + \omega_x H_D\big), \tag{1}$$

with the Cobb–Douglas weights

$$\alpha = (1-\theta_h)(1-\theta_v), \qquad \beta = \theta_h(1-\theta_v), \qquad \gamma = \theta_v, \qquad \alpha + \beta + \gamma = 1.$$

The weights do not depend on addiction status. Referee 2 (C5), Referee 3 (M2a) and the AE (A8) all objected that in the submitted paper the taste weight on drugs rose with d, so reducing SUD mechanically raised the utility derived from c and h and made addiction-reduction and housing expansion complements "even absent incentive effects". Fixing α, β, γ removes that channel completely. It is also a *commitment*: with preferences invariant, becoming addicted no longer relaxes any constraint, so every private cost of SUD must now come from mortality, employment and the budget. Section 7 tests exactly that.

Only the Stone–Geary shifter varies. The AE's suggested alternative (A8) was to vary b instead, and that is what the model does:

$$b_{dmh'} = b_d \;\cdot\; \rho_m^{\mathbf{1}\{m=1\}} \;\cdot\; \rho_H^{\mathbf{1}\{h'=h_1\}}, \qquad d \in \{0,1\}, \quad \rho_m, \rho_H \in (0,1]. \tag{2}$$

A lower b raises the marginal utility of the first units of use. Three separate things are going on here and each is doing a job.

1. b_0 *versus* b_1. A single b cannot match both the participation rate and the drug-expenditure share. A b large enough to deter clean households pushes the unconstrained interior choice v^* below zero, which drives expenditure by households *with* SUD to the corner. Two values separate the moments: b_0 pins participation, b_1 keeps the choice interior so that θ_v pins the expenditure share.

2. $\rho_m < 1$*: self-medication.* Poor mental health raises the marginal value of the first units of use. This is the channel Referee 1 asks about in comment 1(a).

3. $\rho_H < 1$*: the housing-access channel.* This is the only parameter in the model that governs the elasticity of drug use with respect to housing access — the object Referee 2 (C1) calls the crux on which all the welfare results rest, and which Referee 3 and the AE (A13) name independently. Without it that moment is not merely untargeted, it is *untargetable*: housing a household only raises its resources, drugs are a normal good, and the model would predict that housing *raises* use, the opposite sign to Milby et al. (2005). Section 4.2 calibrates ρ_H to that trial.

Leisure. ω_ℓ is the weight on leisure and σ_ℓ its curvature, which pins the Frisch elasticity at $1/\sigma_\ell$.

$\aleph$, the flow value of being alive. $\aleph$ is what makes mortality risk deter drug use: without it, dying is not costly and the model's central mechanism disappears. It also has to satisfy a condition that is easy to overlook — see Section 6.

$\bar{\tau}^u$, the utility cost of treatment. Treatment take-up costs utility, not resources. A resource cost comes out of the same budget that buys food and rent, so the households with the strongest case for treatment — the poorest — would be exactly the ones unable to afford it, which is not how publicly funded treatment works. $\bar{\tau}^u$ is the stigma, effort and time of entering a programme whose monetary cost somebody else bears.

The externality. H is the aggregate homelessness rate and H_D the share of the population both homeless and with SUD. Referee 2's wording is specific — "a large homeless population, *especially if many are addicted to drugs*, creates a negative externality beyond the fiscal burden" — so the disamenity is weighted by composition through ω_x rather than depending on the headcount alone. Households take H and H_D as given, which is what makes this a spillover rather than a private cost, but both are equilibrium objects, so policy moves them and the movement feeds back through the value of life into drug decisions. Section 7.6 calibrates κ_x from a contingent-valuation study rather than leaving it at zero.

2.5 Income, taxes and transfers

Gross income depends on age, employment, productivity and health:

$$y^g(j,y,e,d,m,n) = \begin{cases} 0.40\,y & j \geq j_R \quad \text{(social security)} \\ n\,y\,\varepsilon_j\,\rho_d^{\mathbf{1}\{d=1\}}\rho_m^{\mathbf{1}\{m=1\}} & j < j_R,\ e=1 \quad \text{(employed)} \\ \varrho_{UI}\,y\,\varepsilon_j & j < j_R,\ e=2 \quad \text{(UI, first year only)} \\ \zeta_{DI}\,\bar{y}_{DI}\,\bar{Y} & j < j_R,\ e=3,\ m=1 \quad \text{(disability)} \\ 0 & \text{otherwise,} \end{cases} \tag{3}$$

with, in order of appearance: y the persistent productivity state; ε_j the deterministic age profile of labour efficiency; $\rho_d \leq 1$ and $\rho_m \leq 1$ the productivity penalties from substance use disorder and serious mental illness, applied *conditional on working*; ϱ_{UI} the unemployment-insurance replacement rate, paid on the household's own productivity and only in the first year of a spell; ζ_{DI} the share of the non-employed with serious mental illness who receive disability insurance; $\bar{y}_{DI}$ the disability benefit expressed as a share of average income; and $\bar{Y}$ average income in the economy. Retirement income is a flat 40% replacement of the productivity state, with no health penalty — social security is not means-tested on health.

Why the penalties are conditional on working. This is what allows an *employed* household to have income low enough to choose homelessness. An unemployment effect alone cannot do that, and Referee 1's comment 2(c)(i) is precisely that the model produced a homeless population that was 100% out of work against 40–52% in the data.

Labour income is taxed with the Heathcote–Storesletten–Violante function and capital income at a flat rate:

$$y^n = \tau_\ell\,(y^g)^{\xi} \;+\; (1-\tau_k)\,r\,a. \tag{4}$$

Two transfers. A *food-stamp floor* tops means-tested income up to $\underline{c}\,\bar{Y}$:

$$F = \max\Big\{0,\ \underline{c}\,\bar{Y} - \big(y^g + \max\{0, ra\}\big)\Big\}, \tag{5}$$

and a *housing voucher*, paid to voucher holders and only while renting, covering rent above a tenant contribution of κ_r of gross earnings:

$$Vch = \mathbf{1}\{\eta = 1\}\mathbf{1}\{h' \neq h_1\}\max\big\{0,\ p_h h' - \kappa_r\,y^g\big\}. \tag{6}$$

A homeless voucher holder does not lose the voucher. Retention depends on income eligibility, not on housing, so a household that gives up its unit keeps its entitlement and the voucher simply pays nothing until it rents again. With homeownership gone there is no other tenure for it to be spent on, so this is both the realistic rule and the simple one.

Both are means-tested on *gross* income, as in the programmes themselves. Both are also clawed back — at a rate of one for food stamps and κ_r for the voucher — and that is what generates the welfare trap discussed under labour supply below.

2.6 The budget constraint

$$c + p_v v + a' \;=\; \underbrace{y^n + a}_{\text{after-tax income and assets}} \;-\; \underbrace{p_h h' \mathbf{1}\{h' \neq h_1\}}_{\text{rent}} + Vch + F, \qquad c > 0,\;\; v \geq 0,\;\; a' \geq 0. \tag{7}$$

Note what is *not* in (7): treatment, which costs utility; and bequests, which go to the government.

The closed-form split of c and v. Given a' and h', let R denote the resources remaining. Cobb–Douglas preferences over $(c,\, v+b)$ give

$$c = \frac{\alpha}{\alpha+\gamma}\big(R + p_v b\big), \qquad v = \max\Big\{0,\; \frac{\gamma}{(\alpha+\gamma)p_v}\big(R + p_v b\big) - b\Big\}, \tag{8}$$

so the inner allocation never needs a numerical search. The max is the corner at which the household abstains because the Stone–Geary endowment already exceeds its desired intake — which is exactly how a large b deters use.

2.7 Labour supply

Hours are continuous and satisfy the intratemporal first-order condition

$$u_c(c)\,\frac{\partial R}{\partial n} \;=\; \omega_\ell (1-n)^{-\sigma_\ell}. \tag{9}$$

Two properties of (9) matter substantively.

The residual is not monotone: the welfare trap. From (5) and (6), $\partial R/\partial n$ jumps *up* at the hours where each transfer is exhausted, because the clawback stops. Below the food-stamp exhaustion point the marginal return to an hour of work is close to zero. So "work a little and keep the transfer" and "work properly" are genuinely distinct local optima, and the household compares them. The solution algorithm splits $[0, 1]$ at the kinks and solves each piece exactly; bracketing once would silently pick whichever optimum happened to be bracketed.

Eligibility is a step in the continuation value. Voucher eligibility is lost discretely when means-tested income crosses κ_V times the median. That does not kink the budget — so (9) is continuous across it — but it shifts the continuation value, so the piecewise optimum is clamped to each side of the threshold and both are evaluated.

Referee 3 (m1) wanted labour supply kept and its role made visible; the AE (A3) questioned whether it was needed. The decisive reason to keep it is not the welfare cost of tax financing but the programme itself: **voucher eligibility and the voucher payment are both functions of earnings**, so hours are the margin through which a household controls whether it qualifies and how much it receives. Without an hours choice the voucher's means test would be something that happens to a household rather than something it responds to, and the labour-supply distortion created by a 30% tenant contribution on top of the food-stamp clawback — the welfare trap above — would be invisible. That distortion is a first-order part of what a housing voucher does. Making hours continuous and exact resolves the tension in Referee 3's favour, at a cost of a factor of three in solve time.

2.8 Housing and homelessness

Housing is a four-point grid. h_1 is homelessness, with a small service flow h_{HL} and no rent. The concept is **literal homelessness** — sleeping rough or in an emergency shelter, the HUD point-in-time definition. It *excludes* doubling-up and couch-surfing, which are far more common and would put the rate an order of magnitude higher. The AE's A16 notes that the share of the homeless with SUD depends heavily on which definition is used, and it does: the 21.7% target is measured on the literal-homelessness population, so the model must target the same population. The sensitivity to a broader definition is reported. The three rental sizes are geometric,

$$h_2 = \underline{h}, \qquad h_3 = \underline{h}\,\varsigma, \qquad h_4 = \underline{h}\,\varsigma^2, \tag{10}$$

so two parameters describe the whole grid: the level $\underline{h}$ and the spread ς.

Rent is normalised. Housing services h' enter utility (1); the rent bill $p_h h'$ enters the budget (7). The two are nearly collinear: rescaling the grid by λ and the price by $1/\lambda$ leaves every budget constraint untouched and changes utility only by the factor λ^{β} inside the consumption bundle, which affects behaviour solely through the curvature interaction with the additive leisure and $\aleph$ terms. Identification of the level therefore runs through a weak channel and should not be relied on. Setting $p_h = 1$ makes the *grid* the calibrated object, and $\underline{h}$, ς and h_{HL} are then pinned by three moments of the housing-burden distribution and the homelessness rate. This is what Referee 2's comment D1 asked for: in the submitted paper the grid had seven points, four of them uncalibrated and listed in a footnote.

2.9 Drug use, addiction, recovery and treatment

A household either abstains ($v = 0$) or uses ($v > 0$, chosen continuously according to (8)). Transitions in d:

$$\Pr(d' = 1 \mid d = 0) = \pi_d \cdot \mathbf{1}\{v > 0\}, \qquad \Pr(d' = 0 \mid d = 1) = \pi_r(h', \tau, m). \tag{11}$$

A single addiction hazard. A clean household that uses becomes addicted with probability π_d *regardless of how much it consumes.* There is no light/heavy threshold $\bar{v}$ and no consumption-dependent hazard; the 6.48 heavy-to-light ratio is gone. This removes two of the parameters Referee 3 (M2b) asked to have justified, at the cost of one channel: intensity no longer affects the risk of addiction, only the budget.

Recovery and treatment. The recovery hazard is

$$\pi_r(h', \tau, m) = \min\Big\{1,\ \bar{\pi}_r \cdot \mu_H^{\mathbf{1}\{h' = h_1\}} \cdot \mu_m^{\mathbf{1}\{m=1\}} \cdot g^{\tau}\Big\}, \tag{12}$$

with $\mu_H < 1$ (recovery is harder when homeless), $\mu_m < 1$ (harder with mental illness) and $g > 1$ the treatment effect. $\tau \in \{0, 1\}$ is a choice, available only to households with $d = 1$, at the utility cost $\bar{\tau}^u$ in (1).

Referee 1's comment 1(b) and the AE's A10 are the same point and it is the most damning gap in the submitted paper: the policy is called Treatment First and there was no treatment in the model. There is now, it is a decision rather than a mechanical hazard, and g — through which the entire Treatment First result runs — is disciplined in Section 4.2.

The maximum experimentation age. In the baseline a clean household may initiate only up to $j \leq j_E$. Referee 1's smaller comment 1 notes this is in tension with the U-shaped age profile of use in Table 4(a). It is now a variable rather than a hard-coded parameter, so the model can be run without it; Section 6 reports the age profile both ways.

2.10 The private costs of SUD: mortality, employment, and the homelessness loop

This subsection is the heart of the mechanism and it deserves to be stated explicitly rather than buried in a parameter table.

Mortality. Survival is

$$\phi_j(h', d, m) = 1 - \min\Big\{1,\ \big(1 - \bar{\phi}_j\big) \cdot \kappa_{\phi H}^{\mathbf{1}\{h'=h_1\}} \cdot \kappa_{\phi d}^{\mathbf{1}\{d=1\}} \cdot \kappa_{\phi m}^{\mathbf{1}\{m=1\}}\Big\}, \tag{13}$$

where $\bar{\phi}_j$ is the baseline age profile of survival. SUD multiplies the death hazard by $\kappa_{\phi d}$, homelessness by $\kappa_{\phi H}$ and mental illness by $\kappa_{\phi m}$. Because $\aleph > 0$, a higher death hazard is a real cost, and it is larger for households with more remaining life and more to consume — which is the sorting the paper's mechanism relies on.

Employment. Job finding and separation are

$$\pi^{ue} = \min\big\{1,\ \bar{\pi}^{ue}\, \kappa_{ueH}^{\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{\mathbf{1}\{d=1\}} \kappa_{uem}^{\mathbf{1}\{m=1\}}\big\}, \qquad \pi^{eu} = \min\big\{1,\ \bar{\pi}^{eu}\, \kappa_{ueH}^{-\mathbf{1}\{h'=h_1\}} \kappa_{ued}^{-\mathbf{1}\{d=1\}} \kappa_{uem}^{-\mathbf{1}\{m=1\}}\big\}, \tag{14}$$

so each multiplier below one simultaneously makes work harder to find and easier to lose. On separation the household draws UI with probability π_{UI}:

$$e = 1 \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } 1 - \pi^{eu} \\ e' = 2 & \text{w.p. } \pi^{eu}\pi_{UI} \\ e' = 3 & \text{w.p. } \pi^{eu}(1 - \pi_{UI}) \end{cases} \qquad e \in \{2, 3\} \ \longrightarrow\ \begin{cases} e' = 1 & \text{w.p. } \pi^{ue} \\ e' = 3 & \text{w.p. } 1 - \pi^{ue}. \end{cases}$$

The homelessness loop. Homelessness enters (13), (14), (12) and (2): it raises mortality, worsens employment, lowers the recovery hazard, and raises the marginal utility of drug use. Those four together are what make homelessness and SUD mutually reinforcing, and shutting them off jointly is arm 4 of the mechanism ladder.

What SUD does *not* do. In the baseline there is no direct wage penalty conditional on working, $\rho_d = 1$. The AE (A20) asks about direct productivity effects. The robust findings are on the employment margin — roughly 50% more unscheduled absence and 44% higher turnover, captured by κ_{ued} — while the wage penalty conditional on working is contested, with parts of the older literature finding positive effects for casual use. The baseline therefore imposes none and says so; a version calibrated to the SUD income ratio is reported as an alternative.

2.11 Serious mental illness

m is an exogenous two-state Markov chain. It represents *serious* mental illness, not any mental illness: AMI covers roughly 23% of US adults and includes mild anxiety and depression, whereas SMI is about 5.5% and is defined by substantial functional impairment. A state that halves productivity cannot sensibly cover a quarter of the population.

The onset rate is *derived* from the target prevalence p_m and the recovery rate λ_m,

$$\text{onset} = \lambda_m \frac{p_m}{1 - p_m}, \tag{15}$$

so calibrating persistence never disturbs prevalence.

SMI affects six objects: productivity conditional on working (ρ_m), job finding and separation (κ_{uem}), mortality ($\kappa_{\phi m}$), the recovery hazard (μ_m), the Stone–Geary shifter (ρ_m in (2)), and eligibility for disability income (ζ_{DI} in (3)).

And causation runs the other way too: SUD raises the hazard of SMI. The onset rate is multiplied by $\iota > 1$ for a household with $d = 1$:

$$\Pr(m' = 1 \mid m = 0, d) = \text{onset}_m \cdot \iota^{\mathbf{1}\{d=1\}}. \tag{16}$$

The evidence is strong. Substance-induced psychosis converts to schizophrenia or bipolar disorder at a pooled rate near 25% (Murrie et al., *Schizophrenia Bulletin*), and in the Danish registers any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, amphetamine use disorder 1.90 and opioid use disorder 2.74. The central value is $\iota = 2$ with a range of 1.3–2.7.

This channel matters twice over, and the submitted paper had neither half of it. *Privately* it is a further cost of addiction, so it belongs in the mechanism ladder alongside mortality and employment. *Fiscally* it is a further externality, and a compounding one: SUD raises SMI, SMI raises disability receipt, lowers employment, raises mortality and raises homelessness — so the chain SUD → SMI → DI and lost tax runs entirely outside what the household internalises beyond its own utility. It is measured in §7.6.

Because onset is no longer common to everyone, it can no longer be derived from prevalence: the base rate is calibrated so that population prevalence still lands on 5.5% while the multiplier operates.

The specification follows İmrohoroğlu and Zhao (2022) structurally, not just numerically: in their model labour-market shocks generate *short* homeless spells while health shocks generate *chronic* homelessness, and expanded means-tested disability income is the effective policy for the chronic group. Homelessness driven purely by income risk is transitory; the persistent tail needs a persistent state. This is Referee 1's comment 1(a), and the persistence is also what generates the older homeless population of comment 2(c)(ii).

2.12 The voucher programme and the two policies

A household without a voucher draws one with probability π_V if it is income eligible; a household with one keeps it while eligible. Eligibility is

$$y^g + \max\{0, ra\} \;<\; \kappa_V \cdot \text{median income}. \tag{17}$$

Under **Housing First** that is the whole rule. Under **Treatment First** a voucher is additionally lost with probability p_{det} if the household used drugs, and with probability p_{fp} even if it did not:

$$\Pr(\eta' = 1) = \Pr(\eta' = 1)^{HF} \times \begin{cases} 1 - p_{\text{det}} & \text{if } v > 0 \\ 1 - p_{\text{fp}} & \text{if } v = 0. \end{cases} \tag{18}$$

Referee 1 (4a, 4b) and the AE (A11) both note that perfect observation and enforcement are not credible. p_{det} and p_{fp} are experiment parameters reported as sensitivity, not calibrated, because that is the point.

2.13 The household problem

Let $s = (a, y, e, d, m, \eta)$. The value function solves

$$V_j(s) = \max_{\{n,\, h',\, v,\, \tau,\, a'\}} \left\{ \frac{\left[c^{\alpha}(h')^{\beta}(v + b_{dmh'})^{\gamma}\right]^{1-\sigma}}{1-\sigma} + \omega_{\ell}\frac{(1-n)^{1-\sigma_{\ell}}}{1-\sigma_{\ell}} + \aleph - \tau\bar{\tau}^{u} - \kappa_x\big(H + \omega_x H_D\big) \right.$$
$$\left. + \beta\, \phi_j(h', d, m)\, \mathbb{E}\Big[V_{j+1}\big(a', y', e', d', m', \eta'\big) \;\Big|\; s, h', v, \tau\Big] \right\} \tag{19}$$

subject to (3)–(7), with $V_{J+1} \equiv 0$: there is no bequest motive, and death is worth zero. The expectation is taken over the independent transitions in y (Rouwenhorst), e (14), d (11), m, and η.

The problem is solved by exact backward induction. Within a drug regime and a housing choice the c–v split is closed form (8) and hours come from (9), so the only search is over a', done on the full grid because the objective is *not* single-peaked in a': voucher eligibility is a step function of income (which includes ra), the food-stamp floor kinks, and housing is discrete.

2.14 Government

The government pays social security, UI, disability, food stamps and vouchers, and bears a fiscal cost χ_H per homeless household and χ_D per household with SUD. It finances this with the labour income tax (4), the capital income tax, and the estates of the deceased. The level parameter τ_{ℓ} clears the budget:

$$\underbrace{\sum y^{g} - \tau_{\ell}\sum (y^{g})^{\xi} + \tau_k \sum \max\{0, ra\} + \text{estates}}_{\text{revenue}} = \underbrace{SS + UI + DI + F + Vch + \chi_H H + \chi_D D + G}_{\text{outlay}}. \tag{20}$$

Estates go to the government. Redistributing accidental bequests as a uniform lump sum is a large unconditional transfer to the poor with welfare consequences of its own, which would contaminate every policy comparison in the paper. Sending them to the budget also removes an equilibrium fixed point. The lump-sum version remains available for a robustness run. This is a change from the submitted paper, where transfers were additionally type-dependent — the AE's A7.

2.15 Drug supply and the price

Drug production has cost $C(V) = \frac{\psi_1}{\psi_2} V^{\psi_2}$, so supply satisfies $p_v = \psi_1 V^{\psi_2 - 1}$ and the price elasticity of supply is $1/(\psi_2 - 1)$. The shifter ψ_1 is normalised so that $p_v = 1$ at the benchmark quantity.

This answers Referee 1 (1c), Referee 3 (M2d) and the AE (A5). Referee 3 worried that endogenising the price "requires suppressing this normalization". It does not: the normalisation moves to ψ_1, the benchmark price is still one, and only counterfactuals move it.

What the evidence says about ψ_2, and why it favours the submitted paper. The maintained assumption in the drug-economics literature is that US supply of cocaine and heroin is *very* elastic. Rhodes et al. (2002), discussed in the National Academies' *Understanding the Demand for Illegal Drugs*, assume it outright on the grounds that these are agricultural products with simple, inexpensive processing, so producers can meet any demand at roughly the current price; the reduced-form demand estimates that literature rests on require it.

There is also a natural experiment. In August 2010 Purdue replaced OxyContin with an abuse-deterrent formulation, pushing a very large mass of demand into heroin — enough to quadruple heroin death rates (Alpert, Powell and Pacula, NBER w24475). Purity-adjusted heroin prices did not rise. They fell slightly between 2010 and 2012. A demand shift of that magnitude leaving price flat is direct evidence of a near-horizontal supply curve.

So the baseline here is a *flat* supply curve and finite elasticities are the robustness direction, not the other way round. That reframes the answer to Referee 1 (1c) and Referee 3 (M2d): the submitted paper's fixed price was the field's standard assumption rather than an unjustified normalisation, and the machinery now exists to demonstrate that relaxing it does not matter.

It is worth recording the companion fact, because it disciplines the second stage. The *demand* elasticity is well estimated: participation elasticities of -0.36 to -0.55 for cocaine and -0.80 to -0.90 for heroin (Saffer and Chaloupka), with conditional heroin demand near -0.80 (Olmstead et al. 2015). The paper's -0.63 target sits inside that range, and the -0.36 to -0.9 robustness band Referee 1 asks for *is* that range.

2.16 Stationary equilibrium

A stationary equilibrium is a value function, policy functions, a distribution Φ, and $(\tau_\ell, p_v, \bar{Y}, \text{median}, H, H_D)$ such that households solve (19); Φ is the invariant distribution induced by the policies and the transitions; the government budget (20) holds; the drug market clears at p_v; and $\bar{Y}$, the median, H and H_D are the aggregates implied by Φ.

Note that H and H_D enter (1), so they are genuine fixed points whenever $\kappa_x > 0$, and that the median enters voucher eligibility, so it is a fixed point too — in a skewed income distribution the median sits well below the mean, and using the mean would materially overstate the eligible population.

3 Parameters

Every parameter in the model. **Status** is one of: **N** normalisation or definition; **L** standard in the literature, cited; **D** set directly from data; **C** internally calibrated, with the moment it is assigned

to; **E** determined in equilibrium; **S** reported as a sensitivity rather than calibrated, because no credible point estimate exists.

Table 2: Preferences and prices

Symbol	Meaning	St.	Value, source or target
σ	risk aversion	L	2.0. Attanasio and Weber (2010) survey; the standard life-cycle value
σ_ℓ	curvature on leisure	L	3.0; Frisch elasticity 0.5, Chetty (2012)
ω_ℓ	weight on leisure	C	$\rightarrow$ average hours worked, 40% of time. Guvenen et al. (2023)
θ_h	Cobb–Douglas weight on housing	C	$\rightarrow$ aggregate housing cost / income, 21.2%. SCF 2022, ages 26–85, all households, owner housing at imputed rent; `data/scf_housing_burden.py`
θ_v	Cobb–Douglas weight on drugs	C	$\rightarrow$ drug spending / income given SUD, 44%. Carried over from the submitted paper; its construction must be stated explicitly in the appendix (R1 2e)
b_0	Stone–Geary shifter, clean	C	$\rightarrow$ drug participation, 5.13%. NSDUH 2024 microdata, ages 26+; `data/nsduh_gradients.py`
b_1	Stone–Geary shifter, SUD	C	$\rightarrow$ abstinence rate among homeless adults with SUD, 18.7%. Milby et al. (2005), NO-HOUSING arm: 4.48 consecutive weeks abstinent out of 24. b_1 governs how costly abstention is once a household already has SUD, so one arm of the trial pins the level and the contrast between arms pins the treatment effect. It previously had no target at all, which also left the housing-access ratio undefined — with abstention impossible, the NH arm abstains 0% and the ratio is 0/0
ρ_m	b multiplier, mental illness	C	$\rightarrow$ SUD rate given SMI, 9.75%. NSDUH 2023 detailed tables, ages 26+, by Bayes on Tables 6.13B, 6.4B and 5.10B

Symbol	Meaning	St.	Value, source or target
ρ_H	b multiplier, homelessness	C^2	→ housed/unhoused abstinence ratio, 1.473. Milby et al. (2005), *AJPH* 95(7):1259–65, NACH versus NH arms
ℵ	flow value of being alive	C	→ **model-implied VSL, $7.5m** (Viscusi and Aldy 2003; EPA 2010). It used to target the SUD income ratio, which pins ℵ's effect on WHO uses but not its LEVEL — and at the resulting level 64.6% of the population had $V < 0$, i.e. preferred death, which turns the mortality cost of drug use into a benefit. The SUD income ratio moved to ρ_d
$\bar{\tau}^u$	utility cost of treatment	C	→ treatment take-up given SUD, ≈24%. NSDUH 2023: about a quarter of people with a past-year SUD received any substance use treatment. **Verify against the detailed tables**
β	discount factor	C	→ wealth / GDP = 3. BEA fixed assets against GDP; the standard US private-capital-to-output ratio
p_v	drug price	E	clears the drug market; = 1 at the benchmark by the normalisation of ψ_1
r	risk-free rate	L	0.03, Kaplan, Mitman and Violante (2020)
κ_x	externality weight	D	Inverted from Loubière et al.'s eight-country contingent valuation, €57 per adult per year to end homelessness, normalised per unit of homelessness by the European reference rate. §7.6

Symbol	Meaning	St.	Value, source or target
ω_x	extra weight on homeless-with-SUD	S	**1.0 in the baseline**: the disamenity depends on the headcount alone. No direct estimate exists — R2's wording ("especially if many are addicted to drugs") implies a positive weight and the hedonic split between supportive housing and emergency shelter implies composition matters, but neither pins a number — so the composition-weighted version is an **appendix robustness**, swept over 1–4

Table 3: Housing, taxes and transfers

Symbol	Meaning	St.	Value, source or target
h_{HL}	service flow of homelessness	C	$\rightarrow$ homelessness rate, 0.30%. HUD point-in-time count; *literal* homelessness, excluding doubling-up (AE A16)
$\underline{h}$	smallest rental size	C	$\rightarrow$ share with housing cost $> 50\%$ of income, 17.49%. SCF 2022; `data/scf_housing_burden.py`

Symbol	Meaning	St.	Value, source or target
ς	geometric spread of rental sizes	C	$\rightarrow$ share with housing cost $> 30\%$ of income, 40.62% (SCF 2022). **Better alternative**: target the size distribution directly, from ACS **B25042** (*Tenure by Bedrooms*, universe *occupied* housing units). Its total column is all households by bedrooms, which is the right universe — the model has no homeownership, so every housed household maps to an occupied unit, not to a renter-occupied one. **Not B25041**: that tables universe is *housing units*, which includes the roughly 10% vacant, skewed small and rental, so it overstates the small end. `data/make_housing_dist.py` does the collapse to the models three housed sizes and refuses to do it silently from the wrong universe; the B25042 extract is not in `data/` yet
ξ	tax progressivity (HSV)	L	0.85, Heathcote, Storesletten and Violante (2017)
τ_ℓ	tax level	E	clears the government budget (20) given the distribution
τ_k	capital income tax	L	0.15. US long-run capital-gains and qualified-dividend rate for the middle brackets
$\underline{c}$	food-stamp consumption floor	C	$\rightarrow$ food stamps / labour income, 0.42%. USDA Food and Nutrition Service, SNAP outlays FY2023, against GDP
κ_V	voucher eligibility threshold	D	0.50 of median income, HUD rule
κ_r	tenant contribution rate	D	0.30, HUD rule
π_V	voucher lottery probability	C	$\rightarrow$ voucher recipients / income-eligible, 26.6%. HUD Picture of Subsidized Households against the income-eligible renter population

Symbol	Meaning	St.	Value, source or target
p_{test}	probability a voucher holder is tested, TF	S	1.0 at the benchmark, swept downward. Detection now requires a test, so effective detection is $p_{\text{test}}p_{\text{det}}$ and the old behaviour is the $p_{\text{test}} = 1$ corner
c_{test}	cost per voucher holder tested	D	0.0023 of average income. A urine panel runs about $40 and quarterly testing is roughly $160 a year against average earnings near $70,000. It enters the GOVERNMENT BUDGET, where it competes with everything else the government pays for — R1's comment 4(c) is that the $1–2bn estimate is too large to leave in a dismissive footnote
p_{det}	detection probability given a test	S	1.0 in the benchmark, swept downward. No estimate exists and none should: R1 4(a) and AE A11 object precisely to assuming perfect enforcement, so the point is the sweep
p_{fp}	false-positive rate under TF	S	0.0 in the benchmark, swept upward. As above, R1 4(b)

Symbol	Meaning	St.	Value, source or target
χ_H	fiscal cost per homeless household	D/S	0.304. Abramson and Van Nieuwerburgh (2024, NBER w32582, p. 30) assume \$2,500 per household per month for the *literally* homeless — those in shelters and on the streets, 0.6% of the US population — and half that for the doubled-up homeless (0.83%), giving a population-weighted \$1,775 per month, or \$21,300 a year. Against the model's \$70,000 average income that is $\chi_H = 0.304$; on the literally-homeless figure alone it is 0.429. Their footnote 11 lists the components as shelters, social services, extra policing and hospital emergency care, and they call the blended figure *conservative*. **Note the direction: R2 and R3 both judged the old 0.252 too high, and the best structural estimate in the literature is 20–70% HIGHER, not lower.** It is an assumption inside their calibration rather than an independent estimate, so it is not decisive — but it is the right thing to cite, and it cuts against the referees
χ_D	fiscal cost per household with SUD	D	0.212; CEA (2017), ≈\$30,000 per person-year
G	other public spending	E	Residual that balances (20) at the benchmark, then held fixed across counterfactuals so policy is financed by τ_ℓ rather than by cutting other spending

Table 4: Income process, employment and mortality

Symbol	Meaning	St.	Value, source or target
ρ_y, σ_y	AR(1) in log productivity	L	0.90, 0.20. Storesletten, Telmer and Yaron (2004); Floden and Lindé (2001)
ε_j	deterministic age profile	D	quadratic in age, fitted to the PSID/CPS hump. **Replace with a directly estimated profile**
$\bar{\phi}_j$	baseline survival	D	**Done.** SSA period life table (`oact/STATS/table4c6`), ages 26–85, the two sexes combined by survivor weight $q(x) = [l^M q^M + l^F q^F]/[l^M + l^F]$ rather than averaged equally — by the mid-eighties the survivor populations differ enough that equal weights are simply wrong. Loaded from `data/us_life_table.csv`. The placeholder it replaces understated mortality badly: q at 85 was 0.045 against an actual 0.081, and at 26 0.00054 against 0.00116. This is R3 M1b
$\bar{\pi}^{ue}$	job finding, annual	L	0.55, Krueger, Cramer and Cho (2014)
$\bar{\pi}^{eu}$	job separation	C	$\rightarrow$ unemployment rate, 4.9%. BLS, ages 25+, average over the recent expansion
π_{UI}	share of entrants drawing UI	D	0.30; insured unemployment / total unemployment, FRED 2014–24
ϱ_{UI}	UI replacement rate	D	0.50. DOL statutory replacement rates average close to half of prior weekly wages
κ_{ueH}	employment multiplier, homeless	L	0.80, Desmond and Gershenson (2016)
κ_{ued}	employment multiplier, SUD	C	$\rightarrow$ unemployment rate given SUD, 26.9% (NSDUH 2024, ages 26+, own tabulation in `data/nsduh_gradients.py`). Direction corroborated by roughly 50% more unscheduled absence and 44% higher turnover in the employer literature

Symbol	Meaning	St.	Value, source or target
κ_{uem}	employment multiplier, SMI	C	→ unemployment rate given SMI, 7.1%. NSDUH Table 6.5B: SMI prevalence 8.9% among the unemployed against 5.1% among the full-time employed, inverted by Bayes
$\kappa_{\phi H}$	mortality multiplier, homeless	D	1.60; Meyer et al. (2025), 60% excess mortality
$\kappa_{\phi d}$	mortality multiplier, SUD	C[2]	→ drug-price participation elasticity, −0.63 (inside the −0.36 to −0.55 cocaine and −0.80 to −0.90 heroin participation range of Saffer and Chaloupka); bracketed by the 2.5–9.8 SMR range of Degenhardt et al. (2011)
$\kappa_{\phi m}$	mortality multiplier, SMI	D	2.50. Walker, McGee and Druss (2015), meta-analysis of 203 studies: pooled all-cause mortality risk ratio 2.22, and 10–20 years of life expectancy lost
ρ_d	SUD productivity penalty	D	1.00 in the baseline: no direct wage effect. The employment-margin evidence is robust (about 50% more unscheduled absence, 44% higher turnover) and is carried by κ_{ued}; the wage penalty conditional on working is contested, with parts of the older literature finding positive effects for casual use. A version calibrated to the SUD income ratio is an **appendix robustness** (AE A20)
ρ_m	SMI productivity penalty	C	→ homeless income / average, 19.1%. Meyer et al. (2025), homeless population linked to administrative earnings records

Table 5: Addiction, recovery, treatment and mental illness

Symbol	Meaning	St.	Value, source or target
π_d	addiction hazard given any use	C	→ SUD rate, 3.68%. NSDUH 2024 microdata, ages 26+

Symbol	Meaning	St.	Value, source or target
j_E	maximum experimentation age	S	Real age 55, with no source — R1's smaller comment 1 says so. Run with and without (§6); if it can be dropped it should be
$\bar{\pi}_r$	baseline recovery hazard, housed	D	0.025/yr; Vaillant (1995). The AE questions applying alcohol recovery rates to drug SUD; Fleury et al. (2016) give 35–54% remission at a mean 17-year follow-up across substances, which brackets this
μ_H	recovery multiplier, homeless	C	→ drug use disorder share of the homeless, 21.7%. Gutwinski et al. (2021), *PLOS Medicine*, 39 pooled studies. Note this is the DRUG figure; the submitted paper's 37% is their ALCOHOL figure
μ_m	recovery multiplier, SMI	D	**0.82.** Dual-diagnosis cohorts relapse at 39.8% against 26.4% without a co-occurring disorder (odds ratio 1.85 for difficulty maintaining abstinence), so sustained remission runs 0.602 against 0.736, a ratio of 0.82; converting the odds ratio to a risk ratio at that base rate gives the same figure. Previously 0.50, which was invented
g	treatment effect on recovery	S	1.35, from an odds ratio of 1.22 for contingency management in the only clean randomised evidence. Specialty-treatment remission of 47.6% and overall remission of 35–54% are *not* treatment effects — different populations, different selection — which is why they are not used here. **The TF result runs through this** — report a range and the value at which the ranking flips
p_m	SMI prevalence	C	→ 5.5%, NSDUH 2023, adults. A target rather than a parameter now, since onset varies with SUD

Symbol	Meaning	St.	Value, source or target
λ_m	SMI recovery rate	D	0.067/yr; İmrohoroğlu–Zhao's own moment, ≈50% improve enough to work within 10 years
onset_m	SMI onset	C	→ SMI prevalence, 5.5%. No longer derived, because onset is no longer common to everyone
ι	SUD multiplier on SMI onset	D	**2.0, new.** Danish registers: any substance use disorder predicts conversion to schizophrenia with a hazard ratio of 1.34, amphetamine use disorder 1.90, opioid use disorder 2.74; substance-induced psychosis converts at a pooled 25% (Murrie et al.). Range 1.3–2.7
ζ_{DI}	disability coverage given SMI	C	→ SMI share of the homeless, 25%. Gutwinski et al. (2021): 12.4% schizophrenia-spectrum plus 12.6% major depression. This is İmrohoroğlu–Zhao's own identifying margin (ζ_{DI} in their notation)
$\bar{y}_{DI}$	disability benefit level	D	0.25 of average income. SSA: average SSDI benefit $1,537 a month in 2024, about $18,400 a year against average annual earnings near $70,000, so 0.26. The SSI federal benefit rate of $943 a month is 0.16; SSDI is the relevant figure for people with a work history
ψ_1	drug supply shifter	N	Normalised so $p_v = 1$ at the benchmark quantity, which is what lets the price be endogenous without disturbing the existing normalisation (R3 M2d)

Symbol	Meaning	St.	Value, source or target
ψ_2	drug supply curvature	L	Elasticity $1/(\psi_2 - 1)$, near-horizontal at the baseline. **This is a local claim, not a global one**: supply is not unlimited at a fixed price, and nothing here asserts it is. The claim is that over the range of demand variation these counterfactuals produce, the supply curve is close to flat — which is what Rhodes et al. (2002) assume via the National Academies for cocaine and heroin, and what the 2010 OxyContin reformulation demonstrated, a demand shift large enough to quadruple heroin deaths leaving purity-adjusted prices slightly lower. Outside that range the assumption has no support and none is claimed. Finite elasticities swept as robustness

Superscript 2 marks the two parameters calibrated in the *second* stage, against experimental rather than cross-sectional moments.

Parameters that remain undisciplined. Five, and they are flagged rather than hidden: g (treatment effect), μ_m, $\bar{y}_{DI}$, ψ_2 and κ_x. Of these only g carries a headline result, and it is reported as a range. The baseline survival profile $\bar{\phi}_j$ is currently a smooth functional form and should be replaced with US life tables before the paper is resubmitted.

4 Calibration

Two stages, and they are different exercises. The first matches a cross-section of the stationary equilibrium. The second matches short-run *responses* measured in experiments, which cannot be computed by comparing stationary equilibria.

4.1 First stage: the stationary equilibrium

Nineteen parameters are calibrated jointly against twenty-one moments of the stationary distribution. All the assignments are in Section 3; what matters methodologically is the following.

The parameter–moment map is close to triangular, and the code says so. Each parameter has one moment it dominates, and the sign of $\partial M/\partial\theta$ is declared explicitly for each pair, so a mistaken sign shows up immediately as a moment walking away from its target rather than as a silent failure.

Three moments are step-like and must be bracketed, not stepped. Participation is a near-threshold function of b_0 — below a cutoff everyone uses, above it nobody does, and in between every drug moment reads zero. Homelessness moved from 0.12% to 3.29% for a 35% change in h_{HL}. Wealth/GDP is 5.46 at $\beta = 0.995$ and 0.38 at $\beta = 0.85$. A proportional step of 0.9 takes a discount factor from 0.96 to 0.864, which is enormous in a life-cycle model; the first attempt at this calibration drove β into its lower bound and took every other moment with it. All three are bisected at the head of each sweep, in partial equilibrium, before anything else moves.

The discount factor and wealth/GDP. β targets aggregate wealth equal to three times GDP, with GDP measured as *all factor income* — labour earnings plus the return on the capital stock — and transfers excluded, because transfers are redistribution rather than output. Three is the standard US ratio of the private capital stock to annual output. It is the right target here rather than the SCF's net-worth-to-*household-income* ratio of roughly 7.3, whose numerator is mostly housing equity a model without homeownership cannot hold and whose denominator is not output. One implication should be stated in the paper rather than left for a reader to find: with $A = 3 \cdot \text{GDP}$ and $\text{GDP} = \text{labour income} + rA$, capital income is 9% of GDP, low against a one-third capital share. That follows from holding r fixed at 3% in a model with no production function.

Three targets that were wrong in the submitted paper.

1. *SUD among the homeless.* The paper targets 37%. Gutwinski et al. (2021) pool 39 studies of homeless people in high-income countries and report 36.7% *alcohol* use disorder and 21.7% *drug* use disorder. The model is about drug use. This is the AE's A16.

2. *Income of drug users.* Referee 1 (2c-i) attacked the model's 41% against "roughly 74%". The 74% is computed over people with a use *disorder*, not people who use. For ages 26+, NSDUH gives 95.1% for users and 68.7% for moderate or severe use disorder. Targeting the disorder ratio and *validating* on the use ratio is the sharper test, and it makes the economics explicit: use is close to income-neutral, and it is progression to disorder that concentrates at the bottom.

3. *Housing burden.* Everyone rents in this model, so a renters-only target would import the selection into who rents. The target is the aggregate ratio over *all* households with owner housing valued at imputed rent, and the full distribution is used for validation.

4.2 Second stage: short-run experimental moments

Method note. These are partial-equilibrium **cohort** experiments, not transition paths. A trial of two hundred people does not move the tax rate or the drug price, so prices are held at their benchmark values; the household problem *is* re-solved, because each arm changes the rules a

forward-looking household faces. Only the aggregate policy counterfactuals of R1 3(d) need a transition solver.

Referee 2 (C1): "all of the welfare results rest on whether the model has an accurate elasticity of drug use to housing access". Referee 3 and the AE (A13) say the same independently. Referee 1 (2d, 2e) asks for policy-relevant elasticities from the RCT literature and notes that the price-elasticity moment as written is not replicable.

These are short-run objects. A trial hands a treatment to a selected group and measures them months later; it does not move aggregate prices and it does not wait for a new stationary distribution. The second stage holds prices at benchmark, takes the benchmark distribution as the initial condition, restricts to the trial's eligible population, applies each arm's rules, re-solves the household problem, and pushes the cohort forward.

The housing-access elasticity. Milby et al. (2005) randomised 196 cocaine-dependent homeless adults into three arms, all receiving identical day treatment:

Arm	Model counterpart	n	Weeks abstinent (of 24)
NH	no assistance	66	4.48
NACH	voucher, unconditional — *Housing First*	67	6.60
ACH	voucher, conditional — *Treatment First*	63	8.32

The targeted moment is the *ratio* NACH/NH = 1.473, which identifies ρ_H. Ratios rather than levels: "mean consecutive weeks abstinent out of 24" is not the same object as "abstains this model year", but the treatment effect is comparable and it is the treatment effect the referees want reproduced.

The conditionality effect is a validation check, not a target. The abstract reports that the abstinence-contingent arm beat the unconditional arm, but only *after controlling for treatment attendance*. In the raw randomised comparison NACH versus ACH gives $P = .091$, and the GEE models state the two housed groups "did not differ from one another". Calibrating to a point estimate the source reports as insignificant is exactly the error Referee 1 attacked κ_{cons} for.

The extensive/intensive contrast is the sharper test. Milby's counterfactual is literally no housing, and the effect is large. The Housing First RCTs compare Housing First to "usual services", which already include shelter or transitional housing, and the systematic review finds "no clear differences ... on outcomes related to mental health, quality of life and substance use". These are not in conflict: the elasticity is *large at the extensive margin* and *near zero at the intensive margin*. A model reproducing both is far better identified than one matching either alone.

The drug-price participation elasticity. Perturb p_v, hold everything else at benchmark, re-solve, read the arc elasticity of participation. Partial equilibrium deliberately, since the empirical estimates come from price variation rather than from a comparison of steady states. Target -0.63; identifies $\kappa_{\phi d}$ within the Degenhardt SMR range. Referee 1 also asks for robustness across -0.36 to -0.9, which is the same bisection at the two endpoints.

The value of a statistical life. Reported rather than targeted, because ℵ currently carries a first-stage moment. Reference $7.5m (Viscusi and Aldy 2003; EPA 2010). If the model-implied value is far off, the right response is to move ℵ to the VSL target and reassign the SUD income ratio to ρ_d.

5 Validation

None of these is targeted. Referee 1's 2(a) — model fit never reported — and the AE's A12 are answered by reporting all of them, including the misses, and by framing the misses honestly rather than favourably (2b).

Table 6: Untargeted moments

Moment	Data	Source
Distributional		
Full CDF of housing cost / income	12 thresholds, 5–100%	SCF 2022, ages 26–85, owner housing at imputed rent
Income of drug *users* / average	95.1%	NSDUH 2024, ages 26+
Income of those with SUD / average	68.7% (targeted; reported alongside)	NSDUH 2024, ages 26+
Share of the homeless employed	40–52%	R1 comment 2(c)(i)
Age distribution of the homeless	48.1% aged 26–44; 27.1% 55+	R1 comment 2(c)(ii); to be reported in finer bins
Drug use by age	U-shaped	NSDUH microdata
Wealth by age	—	SCF 2022
Policy responses		
HF effect on 3-year mortality	−4.6pp (s.e. 1.7)	Costantini (2025), ≈300,000 VA veterans, IV
TF effect on long-run health	≈ 0	Costantini (2025). The outcome is mortality and health-service use; the model captures mortality and SUD status, not general morbidity
Housing stability, HF vs TF	sustained vs declining	HF RCT literature
PSH vs emergency shelter, property values	PSH raises, shelter lowers	shelter-siting hedonic literature

The Costantini result is the paper's largest single risk and belongs in the introduction. The best available quasi-experimental evidence says the mortality benefit runs through *housing*, and that conditioning on treatment adds nothing detectable. The submitted paper concludes universal Treatment First dominates. The defensible position after the revision is the *generosity* result — expanding coverage dominates, conditionality is second-order and uncertain — which is

consistent with all of the above and is still a paper. Getting there requires reporting welfare and mortality as two dimensions, the AE's A9, rather than collapsing them into one ranking.

6 Diagnostics the model must pass

Two checks are run before any number is read as a result.

Is life worth living? Death is worth zero in (19). If any household has $V_j(s) < 0$ it strictly prefers to die, and for that household drug use is *rewarded* by the mortality it causes — inverting the paper's central mechanism for exactly the population the policy targets. The diagnostic reports the minimum V over the benchmark distribution, the mass with $V < 0$, and whether they are disproportionately homeless, addicted or retired. If any mass appears, $\aleph$ is too small.

The age profile of use, with and without j_E. Referee 1's smaller comment 1 says the maximum experimentation age is in tension with the U-shaped age profile of use. The test is whether removing it produces a sensible profile or piles use onto the old, who have the least remaining life to lose. If use does concentrate at old ages, that is informative rather than embarrassing — it says the deterrent is the value of remaining life — but it has to be reported and explained rather than assumed away by a hard age limit.

7 Mechanisms: shutting channels and recalibrating b

7.1 Why this replaces the static appendix

Referee 1's comment 1(d) does not merely ask for intuition, it *specifies the design*: show that the mortality channel is the quantitatively important one by shutting channels off. Comment 1(e) says the current appendix is disorganised, static and overreaches. Referee 2 (C4) wants the drug-use policy function shown by income and wealth. Referee 3 (M2b) asks what patterns the Stone–Geary specification actually generates and where heterogeneity in v comes from; (M2c) asks the sharpest theoretical question in any report — since drug use is rational and forward-looking, where do the externalities justifying intervention arise? The AE (A1) asks for a stylised section because the number of moving parts obscures the economics.

One exercise answers all five, run on the *calibrated* model rather than a separate static toy. That matters: a static model with no assets cannot address Referee 2's question about wealth, and a toy calibrated to nothing cannot establish that a channel is quantitatively important.

7.2 The design

Shut one channel at a time and, in each arm, **recalibrate b_0 so that drug participation returns to its benchmark value**. The recalibration is what makes the comparison informative: without it, removing the mortality penalty simply raises use everywhere and the level change swamps any compositional change. Holding participation fixed separates *sorting* from *level* and yields two statistics per arm:

Statistic	What it measures
How far b_0 had to move	how much deterrence the channel was providing
The income gradient of use	*whom* it deterred

The arms run in partial equilibrium at benchmark prices: the object of interest is the behavioural mechanism, not equilibrium feedback, and holding prices fixed isolates it as well as being far cheaper.

7.3 The arms

1. **Benchmark.** All channels on.
2. **No mortality penalty**, $\kappa_{\phi d} = 1$. The paper's headline claim.
3. **No employment penalty**, $\kappa_{ued} = 1$. Referee 1's rival hypothesis.
4. **No homelessness feedback**, $\kappa_{\phi H} = \kappa_{ueH} = \mu_H = 1$ and $\rho_H = 1$. Isolates the loop that makes homelessness and SUD mutually reinforcing.
5. **All private costs off at once.** Whatever deterrence survives is *resource diversion*: the budget constraint alone. Diversion cannot be switched off directly — drugs are priced inside a Cobb–Douglas bundle, so making them free sends v to the corner — but it is measurable as the residual, which is what this arm reports.

7.4 What is reported

Table 7: Mechanism decomposition (structure; numbers pending the calibration)

	Benchmark	No mortality penalty	No employment penalty	No homeless feedback	All private costs off
b_0 relative to benchmark	1.00				
Participation (%, held fixed)					
SUD rate (%)					
Homelessness (%)					
Use rate, income Q1 (%)					
Use rate, income Q5 (%)					
User income / average					
experimenters only					
SUD only					
Users unemployed (%)					
Users homeless (%)					
Mean age of users					
Mean assets of users					

7.5 The sharp prediction, and how it could fail

With preferences fixed across addiction status, becoming addicted no longer relaxes any constraint, so *all* deterrence must come from mortality, employment and the budget. If removing the mortality penalty requires only a small change in b_0 and leaves the income gradient nearly flat, the paper's central mechanism is not carrying the weight the text claims, and the honest response is to say so and re-centre on the generosity result. If b_0 must fall substantially and the gradient inverts, the mechanism is established numerically on the calibrated model — which is what Referee 1 asked for and what the static appendix could not deliver.

7.6 Where the externalities come from

Referee 3's M2c asks where the externalities justifying intervention arise, given that use is rational and forward-looking. They are of two kinds.

Fiscal, and it has five parts. A household that acquires SUD (i) works less and pays less tax; (ii) becomes likelier to be voucher-eligible and to receive one; (iii) becomes likelier to be homeless, which raises the direct programme cost *and* makes work harder to find and easier to lose ($\kappa_{ueH} = 0.80$, Desmond and Gershenson), compounding back into (i); (iv) becomes likelier to develop serious mental illness, which raises disability receipt and feeds back into employment, mortality and homelessness; and (v) dies earlier, truncating the whole stream of future revenue. None of this is internalised, and all of it is already in the model — but the submitted paper never reported it, and did not have channel (iv) at all.

It is now measured. Define the expected present discounted value of a household's net contribution to the government,

$$N_j(s) = \underbrace{T_j(s)}_{\text{taxes}} - \underbrace{Vch_j(s)}_{\text{vouchers}} - \underbrace{Tr_j(s)}_{\text{other transfers}} - \underbrace{\chi_H \mathbf{1}\{h' = h_1\} + \chi_D \mathbf{1}\{d = 1\}}_{\text{direct costs}} + \frac{\phi_j(s)}{1+r}\,\mathbb{E}\big[N_{j+1}(s')\big], \quad (21)$$

discounted at the government's rate. Given the benchmark policy functions this is a *linear* recursion, so it costs one backward pass and no optimisation, and carrying the four components separately decomposes the answer. The fiscal cost of one more case is

$$\mathbb{E}\big[N(a, y, e, d = 0, m, \eta)\big] - \mathbb{E}\big[N(a, y, e, d = 1, m, \eta)\big]$$

over the benchmark distribution of clean households: the same household, counterfactually with and without SUD.

Two of the four channels are separated by re-running the recursion with one transition altered. Evaluating the SUD stream at *clean* survival isolates what early death alone costs, purely by truncating the revenue stream. Evaluating it with homelessness carrying *no* employment penalty isolates the SUD $\rightarrow$ homelessness $\rightarrow$ unemployment $\rightarrow$ lost-tax chain. Both are easy to miss precisely because they work through transitions rather than through any flow.

Reporting them in one currency. The fiscal loss has to be financed by somebody, so annuitising it and spreading it across the population puts it in the same units as the disamenity — a share of average annual consumption. Only then can a reader see which channel actually matters. On the target values alone the direct outlays $\chi_H H + \chi_D D$ come to about 0.86% of average income per capita against a disamenity near 0.33%, and the direct outlays are only one of the four fiscal

channels. If that ordering survives the calibration it is a substantially stronger position than the submitted paper's, where the entire ranking rested on κ_{cons}: the case for intervention would rest mainly on the channel that is measured inside the model and needs no free parameter, rather than on the contingent-valuation number, which is the weakest link.

A second, wider measure. The present-value recursion above is exact but partial: it prices what the government gains and loses, not what *households* gain and lose, so it misses three things. It misses the deadweight loss of financing the transfer, which is real and first-order here because labour supply is elastic. It misses the drug price, which is an equilibrium object, so one more user raises the price every other user pays. And it misses the transfer scale: average income sets the food-stamp floor and the disability benefit, median income sets voucher eligibility, so one more case of SUD lowers all three for everybody. The wider measure raises the addiction hazard π_d by ten per cent, re-solves, and reports the change in newborn welfare as a consumption-equivalent variation. Each channel is then isolated by re-solving with that channel alone *frozen* at its benchmark value; the gap between the free re-solve and the frozen one is that channel's contribution.

Two cautions on reading the resulting table. First, the total includes the household's own expected cost, which it internalises and which is therefore *not* an externality; only the channels beneath it are. Second, the disamenity, drug-price and transfer-scale channels are each measured with the other channels live, so they are marginal contributions and need not sum to the total — the sum is a diagnostic, not an accounting identity. The fiscal pair is the exception: the mechanical transfer is measured by holding hours at their benchmark schedule while the tax moves, so no distortion is created, and the deadweight loss is the residual once hours are free to respond. Those two are nested by construction and do add up. Holding hours fixed has to be done by constraining the choice, not by reweighting leisure in preferences: a large weight on leisure does not fix hours, it drives them to the corner.

Non-fiscal. The disamenity term in (1). Referee 2 (C2) asked for it and said it was easy to introduce; it is, but it has no credible point estimate. Hedonic shelter-siting studies measure a *local, avoidable* disamenity — what people pay to live further away — and this model has no geography, so nobody can avoid anything and a capitalisation coefficient does not map onto an aggregate public bad. Corbae, Glover and Nattinger's externality is about neighbourhood housing *quality* rather than homelessness, and their identification comes from rent decontrol, so what transfers is their identification logic, not their functional form.

A calibration for the disamenity does exist. Referee 2 noted that introducing the externality "would require a new calibration moment". There is one, and it is a much better fit to this model than the hedonics. Loubière et al. ran an eight-country contingent-valuation study (France, Ireland, Italy, Netherlands, Poland, Portugal, Spain, Sweden, roughly 700 respondents each, fielded in 2017) asking exactly the right question: what would you pay in annual taxes to *end* homelessness through Housing First? The answer was about €57 per respondent per year, roughly 0.2% of average annual earnings. That is a stated preference over the *aggregate* homelessness rate — precisely the object κ_x multiplies — and it is non-spatial, which the hedonic estimates are not.

The normalisation that this requires. The €57 buys elimination of the *European* homelessness rate, not the model's. Dividing by the model's own rate would make the same survey figure imply a different disamenity in every economy it was transplanted into: mechanically small wherever homelessness is rare and large wherever it is common, which is the opposite of what a per-unit valuation means. So κ_x is set as a price per unit of homelessness,

$$\kappa_x = \frac{\text{WTP} \cdot \mathbb{E}\big[c\, u_c\big]}{H_{\text{ref}}\big(1 + \omega_x s_{\text{ref}}\big)}, \tag{22}$$

with H_{ref} the European rate at the time of the survey and s_{ref} the drug-use-disorder share of the homeless in the same pooled evidence base. FEANTSA counted roughly 700,000 people sleeping rough or in shelter across the EU in 2017 against a population near 440 million, so $H_{\text{ref}} \approx 0.16\%$; the 2023 overview reports about 895,000, but part of that rise is improved measurement. The central value used is 0.18% and the range swept is 0.15–0.22%. Since κ_x is inversely proportional to H_{ref}, this is the single most consequential judgement in the disamenity calibration and it must be reported as a range. At a model homelessness rate of 0.30% against a reference of 0.18%, the adjustment scales the disamenity by 1.67, from 0.20% to 0.33% of income.

Caveats, to be stated in the paper. Stated rather than revealed preference; European rather than US; 30.3% protest zeros; and the elicited change is elimination rather than a marginal change, so the linear extrapolation to the model's rate is an assumption.

What the hedonics are still good for. Not a magnitude, but a sign. The NYC Independent Budget Office finds a property within 500 feet of a family shelter sells for about 6.4% less than a comparable property 500–1,000 feet away, and properties near two or more shelters 17.4% less than near one. The Furman Center, studying 123 supportive-housing developments over 18 years, finds supportive housing does *not* depress neighbouring values and slightly raises the closest ones; spatial difference-in-differences work reaches the same split. Permanent supportive housing is Housing First-like and temporary congregate shelter is Treatment First-like, so the externality plausibly differs in *sign* between the two policies being compared. That is a substantive point in Housing First's favour and it does not require a number.

Note finally that $\kappa_x = 0$ is a defensible baseline in a way $\kappa_{\text{cons}} = 0$ was not: that was the value at which the paper's conclusion reversed.

8 What each referee still wants

8.1 What has been done, and what it answers

Table 8: Model changes and the comments they answer

#	Change	Answers
1	Homeownership, tenure, mortgages and negative assets removed	AE A4; R2 D1
2	Housing-market GE removed	AE A5

#	Change	Answers
3	Drug price made an equilibrium object, ψ_1 normalised	R1 1(c); R3 M2d; AE A5
4	Cobb–Douglas weights fixed across SUD status	R2 C5; R3 M2a; AE A8
5	κ_{cons} deleted; homelessness externality in period utility, weighted by composition	R1 3(a), 3(b); R2 C2; R3 M2c; AE A9
6	Endogenous treatment take-up, at a utility cost, with a take-up target	R1 1(b); AE A10
7	Imperfect detection and false positives	R1 4(a), 4(b); AE A11
8	Serious mental illness, six channels	R1 1(a), 2(c)(ii)
9	SUD wage penalty set to one in the baseline, with the evidence stated	AE A20
10	UI drawn on entry, one year; four states collapse to three exactly	AE A2
11	Estates to the government rather than lump-sum by type	AE A7
12	Single addiction hazard; $\bar{v}$ and the 6.48 ratio gone	R3 M2b (partly)
13	Continuous labour supply from the FOC, welfare trap handled exactly	R3 m1; AE A3
14	β retargeted to wealth/GDP $= 3$	AE A17
15	Rent normalised; the whole housing grid calibrated	R2 D1
16	Homeless SUD target corrected 37% $\rightarrow$ 21.7%	AE A16
17	Drug-user income target split: users 95.1%, SUD 68.7%	R1 2(c)(i)
18	Housing burden targeted over all households, full CDF for validation	R1 2(a); AE A12
19	Every parameter given a source or a target (§3)	AE A17
20	Capital income taxed in the household budget; true median income	internal consistency
21	Mechanism ladder built	R1 1(d), 1(e); R2 C4; R3 M2b, M2c; AE A1

8.2 Referee 1 — outstanding

#	Comment	Status
1(d),(e)	Show the mortality channel is the quantitatively important one; reorganise the appendix	Ladder built, **not yet run** — needs the calibration
2(a)	Model fit never reported	**Pending the calibration**
2(b)	Tables 3–4 misses framed too positively	**Writing**, once the misses are known
2(c)(i)	Homeless 100% unemployed vs 40–52%; income ≈ 0 vs 19.1%	Mechanism added; the employment share is **not yet a targeted moment**
2(c)(ii)	Homeless too young	SMI added; **age distribution not yet computed**, and needs finer bins
2(d)	Match policy-relevant RCT elasticities	**Second stage built, not yet run**
2(e)	Price-elasticity moment not replicable	Model counterpart defined; **appendix not written**
2(f)	Show identification numerically	**Not started.** One solve per parameter
3(c)	Tone down; universal TF not unambiguously preferred	**Writing**
3(d)	Solve proper transition paths	**Not built.** The critical path
4(c)	Compliance, stigma and administrative costs of testing	**Not started.** Needs a real admin cost in the budget
4(d)	Sensitivity ladder over addiction, recovery, mortality, employment	**Not started**
sm. 1	Why a maximum experimentation age?	Now switchable; **diagnostic not yet run**
sm. 2–10	Levels and percentage changes; typographical and citation fixes	**Writing**

8.3 Referee 2 — outstanding

#	Comment	Status
C1	The elasticity of drug use to housing access	ρ_H added and the moment defined; **not yet calibrated**
C2	Homelessness externality	Implemented, $\kappa_x = 0$; **no credible calibration**, see §7.6

#	Comment	Status
C3	Parental anxiety about children	**Open decision.** R2 says it "could be discussed"; a warm-glow term would answer this and AE A6 together
C4	Show the drug-use policy function by income and wealth	**Not produced.** Free once calibrated
C6	Optimal voucher design; is χ_H too high?	**Not started.** One sweep of π_V plus a χ_H sensitivity

8.4 Referee 3 — outstanding

#	Comment	Status
M1a	Stylised facts on drug use, voucher receipt and homelessness	**Not written**
M1b	Empirical basis for survival and unemployment hazards	Partly done in §3; $\bar{\phi}_j$ is still a placeholder and must come from life tables
M1c	Show the empirical income–drug-use relationship	Microdata extracted; **figure not made**
M2b	What patterns does Stone–Geary generate?	Ladder built, **not run**
M2c	Where do the externalities arise?	Enumerated in §7.6; the fiscal side needs quantifying from the ladder
m1	Labour supply and the welfare cost of tax financing	Labour supply now continuous; **welfare decomposition not done**
m2	Sensitivity as graphs, not tables	**Not started**
m3	Vary voucher receipt continuously; Laffer curve?	**Not started**

8.5 Associate Editor — outstanding

#	Comment	Status
A1	Stylised model section	Plan is to use the calibrated model plus the ladder; **not written**
A6	Intergenerational language without intergenerational linkages	**Open decision**: delete the language, or add a warm-glow term (with R2 C3)

#	Comment	Status
A9	Report welfare and mortality as two dimensions	**Not done.** Requires the counterfactuals
A12	Model fit; non-targeted moments	**Pending the calibration**
A13	Key elasticities	**Second stage built, not run**
A14	Connect to Costantini, Cohen, Brounstein and Wieselthier, Abramson, Humphries et al.	**Writing**
A15	Compare χ_H with Abramson and van Nieuwerburgh	**Done**, §3. Their \$1,775/month blended figure implies $\chi_H = 0.304$ and their literally-homeless \$2,500 implies 0.429, both above the 0.252 the referees called too high
A16	Homelessness definition	Target corrected; the paper must still state that it targets *literal* homelessness and excludes doubling-up
A17	Rewrite the calibration section	§3–4 are the raw material; **not yet in the paper**
A18	Fewer counterfactuals in the body	**Writing**
A19	"Addicted" → "suffering from substance use disorder"	**Writing**, global replace

8.6 Loose ends: what is *not* covered by the work already planned

Almost everything in the four lists above is closed by one of four blocks of work that are already scoped: the **recalibration**, the **counterfactuals**, the **mechanism ladder**, and the **rewriting**. Stripping those out leaves a short list of genuine loose ends — items that need a decision, new data, or code that does not exist yet.

Ref	Loose end	What it actually needs
R1 3(d)	Transition paths	**Written, and now executing.** No longer a loose end in the sense of missing code. What remains is cost: one path iteration is two passes over T dates by 60 ages, so $T = 60$ with thirty iterations would be nineteen hours. $T = 30$ and a better initial path are the mitigations

Ref	Loose end	What it actually needs
R1 4(c)	Administrative cost of drug testing	**Done.** Each voucher holder faces a per-period probability of being tested, each test costs the government, and detection now requires a test. It is in the budget constraint. The remaining judgement is the cost per test, currently $40 quarterly
R2 C3 / AE A6	Parental anxiety; intergenerational language without linkages	**A decision, and here is what it is between.** The submitted paper's §4.1.3 discusses intergenerational transmission but the model has no link between parents and children, which is the AE's A6. R2 separately says parents suffer from the risk that their children become addicted or homeless, and predicts it would amplify the welfare gains. **Option 1**, delete the §4.1.3 language: costs nothing, answers A6, ignores R2. **Option 2**, add a warm-glow term at one parental age — utility loads on the expected state of a child drawn from the $\rho_x = 0.5$ process the paper already specifies: one term, answers both, and turns a criticism into a mechanism. **Recommended: option 2**
R1 sm. 1	Maximum experimentation age	**A diagnostic and possibly a defence.** If dropping it piles use onto the old, that has to be explained rather than assumed away
R2 C2	ω_x, the composition weight	**Genuinely open, and searched for.** What the literature supports is a difference in SIGN by facility type – permanent supportive housing raises nearby property values, temporary emergency shelter lowers them – not a composition weight on the homeless population's addiction rate. Baseline 1.0, swept 1–4, and the sign asymmetry carried as the qualitative point

Ref	Loose end	What it actually needs
R3 M1b	Survival hazards	**Closed.** SSA period life table loaded from `data/us_life_table.csv`; survival is no longer a fitted exponential
AE A16	Homelessness definition	**Done** — see the statement in §2.7 above
R1 2(e)	Replicability of the price-elasticity moment	**An appendix defining each moment's model counterpart.** Mechanical, but nobody has written it
—	g, the treatment effect	**Closed enough.** One clean randomised estimate exists, an odds ratio of 1.22 for contingency management, and that is a real number rather than a gap. It stays a reported range because the TF result runs through it, not because it is unidentified

That is nine items, of which one is a real build (transition paths), two are decisions rather than work, two need data that exists but has not been pulled, two are writing that nobody has started, and two — ω_x and g — are things the evidence does not support pinning down and which should be presented as ranges rather than resolved. Every other comment in §8 is closed by the recalibration, the counterfactuals, the ladder, or the rewrite.

8.7 The critical path

Three things gate almost everything above.

1. **The first-stage calibration converging.** Unblocks R1 2(a), 2(b), 2(c), 1(d), R2 C4, R3 M2b, AE A12 at once.
2. **The second-stage calibration**, which is built and needs only a converged benchmark to run against. This is R2 C1, R3 and AE A13 — what all three independently call the crux — and it is a partial-equilibrium cohort experiment, so it does *not* wait on the transition solver.
3. **The counterfactuals.** Unblock AE A9, A18, R2 C6, R3 m3, R1 3(c), 4(d). Then the transition-path solver for R1 3(d), which is the one large build left and affects only the aggregate policy experiments.

Everything else on the outstanding lists is either writing or a single cheap run.